



Solution Set

Question 1

Consider an equity option under Merton jump diffusion. Define the state variables (diffusion volatility, Poisson intensity, jump mean, jump volatility, and compensator) and derive the jump-compensated risk-neutral SDE. State the measure, filtration, and tradability assumptions.

$$\frac{dS_t}{S_{t-}} = (r - \lambda\kappa)dt + \sigma dW_t + (J - 1)dN_t$$

Answer key

Define the tradable state variables first: diffusion volatility, Poisson intensity, jump mean, jump volatility, and compensator. The payoff is then written at the relevant horizon and valued under the pricing measure.

The derivation uses the economic relation for jump-compensated risk-neutral SDE. If a hedge or replication argument is used, the stochastic exposure is neutralized locally and the remaining cash account earns the risk-free or collateral rate.

The final expression must agree with the payoff at maturity and with the obvious limiting cases.

$$\begin{aligned} \frac{dS_t}{S_{t-}} &= (r - \lambda\kappa)dt + \sigma dW_t + (J - 1)dN_t \\ d\Pi &= dV - \Delta dS, \quad d\Pi = r\Pi dt \end{aligned}$$

Question 2

For an equity option under Merton jump diffusion, explain which probability measure is natural for pricing. Write the density-process logic needed if the starting model is under the physical measure.

Answer key

The pricing measure is chosen so discounted tradable prices are martingales under the chosen numeraire.

The Radon-Nikodym density reweights probabilities; it does not change the contractual payoff. In Brownian models, Girsanov changes the drift and leaves the quadratic variation unchanged.

A useful check is that the density has expectation one and that the resulting discounted price process has zero drift.

$$\begin{aligned} \frac{dS_t}{S_{t-}} &= (r - \lambda\kappa)dt + \sigma dW_t + (J - 1)dN_t \\ Z_t &= \frac{dQ}{dP} \Big|_{\mathcal{F}_t}, \quad E^Q[X | \mathcal{F}_t] = \frac{1}{Z_t} E^P[Z_T X | \mathcal{F}_t] \end{aligned}$$

Question 3

Identify the discounted tradable or density-adjusted process in an equity option under Merton jump diffusion. Prove the martingale condition and name one modelling choice that would break it.

Answer key

The martingale candidate is the discounted tradable price, or the density-adjusted process after a measure change.

Condition on the earlier information set and verify that the conditional expectation equals the earlier value. Any remaining predictable drift indicates a missing funding, dividend, numeraire, or model adjustment.

For this topic, the practical issue is forgetting the compensator and therefore shifting the forward price.

$$\begin{aligned} \frac{dS_t}{S_{t-}} &= (r - \lambda\kappa)dt + \sigma dW_t + (J - 1)dN_t \\ E[M_t | \mathcal{F}_s] &= M_s, \quad 0 \leq s < t \end{aligned}$$

Question 4

Using jump intensity 0.7, mean jump -8 percent, jump volatility 18 percent, and one-year maturity, construct a two-step numerical calculation. Show intermediate state values before discounting or aggregating.

$$\frac{dS_t}{S_t} = (r - \lambda\kappa)dt + \sigma dW_t + (J - 1)dN_t$$

Answer key

Use the stated market inputs: jump intensity 0.7, mean jump -8 percent, jump volatility 18 percent, and one-year maturity. Compute future state values first, then take the correct conditional average, then discount.

Keep a calculation table with states, probabilities, payoffs, discount factors, and final present values. This makes unit mistakes visible.

The answer should include a sign and magnitude check, not only a final number.

$$\begin{aligned}\frac{dS_t}{S_t} &= (r - \lambda\kappa)dt + \sigma dW_t + (J - 1)dN_t \\ V_t &= e^{-r\Delta t}(qV_u + (1 - q)V_d)\end{aligned}$$

Question 5

For an equity option under Merton jump diffusion, compute and interpret diffusion delta versus unhedgeable jump exposure. Explain the hedge sign and the unit of each sensitivity.

Answer key

The relevant exposure is diffusion delta versus unhedgeable jump exposure. Compute the first-order term, then add convexity or cross terms only when the state variables require them.

A hedge has the opposite sign of the exposure. Delta is per unit of underlying, DV01 is per basis point, and vega is per volatility point or percent depending on convention.

After hedging, report the residual risk that remains.

$$\begin{aligned}\frac{dS_t}{S_t} &= (r - \lambda\kappa)dt + \sigma dW_t + (J - 1)dN_t \\ dV &\approx \Delta dS + \frac{1}{2}\Gamma(dS)^2 + \nu d\sigma\end{aligned}$$

Question 6

Suppose the desk observes short-dated smile and crash put quotes. Write the calibration objective, the weights you would use, and the parameter-identifiability checks.

Answer key

The calibration instruments are short-dated smile and crash put quotes. Write a weighted least-squares objective using market quotes and model quotes.

Weights should reflect liquidity, bid-ask width, and quote reliability. Parameter constraints should prevent impossible prices or unstable extrapolation.

After calibration, inspect residuals and check whether each parameter has a distinct effect on the quotes.

$$\begin{aligned}\frac{dS_t}{S_t} &= (r - \lambda\kappa)dt + \sigma dW_t + (J - 1)dN_t \\ \hat{\theta} &= \arg \min_{\theta} \sum_i w_i (M_i(\theta) - Q_i^{\text{mkt}})^2\end{aligned}$$

Question 7

State the terminal and boundary conditions for an equity option under Merton jump diffusion. Explain how compound Poisson distribution and terminal payoff prevents economically impossible prices.

$$\frac{dS_t}{S_t} = (r - \lambda\kappa)dt + \sigma dW_t + (J - 1)dN_t$$

Answer key

The terminal condition comes from the payoff. Boundary conditions come from economic limits such as zero spot, very large spot, barrier touch, exercise, or discount-factor positivity.

For this product, the key boundary idea is compound Poisson distribution and terminal payoff.

A boundary condition is wrong if it permits arbitrage, negative values for non-negative payoffs, or a price that violates obvious monotonicity.

$$\frac{dS_t}{S_t} = (r - \lambda\kappa)dt + \sigma dW_t + (J - 1)dN_t$$
$$V(T, S) = g(S), \quad V(t, S_{\text{boundary}}) = \text{economic boundary value}$$

Question 8

Design a Monte Carlo or lattice estimator for an equity option under Merton jump diffusion. Specify diffusion paths with random jump counts and jump sizes, the estimator, the discounting rule, and the standard error.

Answer key

Simulate diffusion paths with random jump counts and jump sizes. For each path, compute the payoff using only information available along that path.

The estimator is the discounted sample average. Report standard error and rerun with larger path counts or finer time steps to separate sampling error from discretization error.

For path-dependent products, store the running state variable explicitly.

$$\frac{dS_t}{S_t} = (r - \lambda\kappa)dt + \sigma dW_t + (J - 1)dN_t$$
$$\widehat{V}_0 = e^{-rT} \frac{1}{N} \sum_{i=1}^N X^{(i)}, \quad \widehat{\text{SE}} = e^{-rT} \frac{s_X}{\sqrt{N}}$$

Question 9

A junior quant implements an equity option under Merton jump diffusion and gets a stable but wrong answer. Explain how forgetting the compensator and therefore shifting the forward price creates the error and design a diagnostic test.

Answer key

The main implementation risk is forgetting the compensator and therefore shifting the forward price. This can produce stable but wrong results because the error is systematic.

A good diagnostic compares the implementation with a limiting case or benchmark where the direction of the answer is known.

Change one input at a time and verify monotonicity, units, and convergence.

$$\frac{dS_t}{S_t} = (r - \lambda\kappa)dt + \sigma dW_t + (J - 1)dN_t$$
$$\text{error}(N, \Delta t) = \widehat{V}_{N, \Delta t} - V_{\text{benchmark}}$$

Question 10

Write an NumPy algorithm for an equity option under Merton jump diffusion. Include inputs, outputs, diagnostics, and one benchmark case.

$$\frac{dS_t}{S_t} = (r - \lambda\kappa)dt + \sigma dW_t + (J - 1)dN_t$$

Answer key

Create clear arrays for inputs, states, probabilities, discount factors, and payoffs. Check dimensions before combining them.

Return both the main value and diagnostics such as convergence, residuals, standard error, or benchmark difference.

The routine is acceptable only if the result can be reproduced from the displayed assumptions and data.

$$\frac{dS_t}{S_t} = (r - \lambda\kappa)dt + \sigma dW_t + (J - 1)dN_t$$

$$\text{tree cost} = O(n^2), \quad \text{simulation cost} = O(NM)$$

Question 11

Starting from the payoff definition of an equity option under Merton jump diffusion, derive a replication, dynamic-programming, or conditional-expectation representation without skipping the economic argument.

Answer key

Define the tradable state variables first: diffusion volatility, Poisson intensity, jump mean, jump volatility, and compensator. The payoff is then written at the relevant horizon and valued under the pricing measure.

The derivation uses the economic relation for jump-compensated risk-neutral SDE. If a hedge or replication argument is used, the stochastic exposure is neutralized locally and the remaining cash account earns the risk-free or collateral rate.

The final expression must agree with the payoff at maturity and with the obvious limiting cases.

$$\frac{dS_t}{S_t} = (r - \lambda\kappa)dt + \sigma dW_t + (J - 1)dN_t$$

$$d\Pi = dV - \Delta dS, \quad d\Pi = r\Pi dt$$

Question 12

Create a stressed-market version of jump intensity 0.7, mean jump -8 percent, jump volatility 18 percent, and one-year maturity. Recompute the value or risk metric and explain which term dominates the change.

Answer key

Use the stated market inputs: jump intensity 0.7, mean jump -8 percent, jump volatility 18 percent, and one-year maturity. Compute future state values first, then take the correct conditional average, then discount.

Keep a calculation table with states, probabilities, payoffs, discount factors, and final present values. This makes unit mistakes visible.

The answer should include a sign and magnitude check, not only a final number.

$$\frac{dS_t}{S_t} = (r - \lambda\kappa)dt + \sigma dW_t + (J - 1)dN_t$$

$$V_t = e^{-r\Delta t}(qV_u + (1 - q)V_d)$$

Question 13

Use a second-order Taylor approximation for an equity option under Merton jump diffusion. Separate linear exposure, convexity exposure, cross terms if relevant, and residual model risk.

$$\frac{dS_t}{S_t} = (r - \lambda\kappa)dt + \sigma dW_t + (J - 1)dN_t$$

Answer key

The relevant exposure is diffusion delta versus unhedgeable jump exposure. Compute the first-order term, then add convexity or cross terms only when the state variables require them.

A hedge has the opposite sign of the exposure. Delta is per unit of underlying, DV01 is per basis point, and vega is per volatility point or percent depending on convention.

After hedging, report the residual risk that remains.

$$\begin{aligned} \frac{dS_t}{S_t} &= (r - \lambda\kappa)dt + \sigma dW_t + (J - 1)dN_t \\ dV &\approx \Delta dS + \frac{1}{2}\Gamma(dS)^2 + \nu d\sigma \end{aligned}$$

Question 14

Show how a change of numeraire or change of measure would simplify one expectation in an equity option under Merton jump diffusion. Identify what becomes a martingale after the change.

Answer key

The pricing measure is chosen so discounted tradable prices are martingales under the chosen numeraire.

The Radon-Nikodym density reweights probabilities; it does not change the contractual payoff. In Brownian models, Girsanov changes the drift and leaves the quadratic variation unchanged.

A useful check is that the density has expectation one and that the resulting discounted price process has zero drift.

$$\begin{aligned} \frac{dS_t}{S_t} &= (r - \lambda\kappa)dt + \sigma dW_t + (J - 1)dN_t \\ Z_t &= \frac{dQ}{dP} \Big|_{\mathcal{F}_t}, \quad E^Q[X | \mathcal{F}_t] = \frac{1}{Z_t} E^P[Z_T X | \mathcal{F}_t] \end{aligned}$$

Question 15

Give a proof-style answer showing when the wealth process for an equity option under Merton jump diffusion is self-financing. Explain why predictable trading strategies matter.

Answer key

The martingale candidate is the discounted tradable price, or the density-adjusted process after a measure change.

Condition on the earlier information set and verify that the conditional expectation equals the earlier value. Any remaining predictable drift indicates a missing funding, dividend, numeraire, or model adjustment.

For this topic, the practical issue is forgetting the compensator and therefore shifting the forward price.

$$\begin{aligned} \frac{dS_t}{S_t} &= (r - \lambda\kappa)dt + \sigma dW_t + (J - 1)dN_t \\ E[M_t | \mathcal{F}_s] &= M_s, \quad 0 \leq s < t \end{aligned}$$

Question 16

Market quotes for short-dated smile and crash put quotes are noisy. Propose a robust calibration workflow with initial guesses, constraints, residual plots, and out-of-sample checks.

$$\frac{dS_t}{S_t} = (r - \lambda\kappa)dt + \sigma dW_t + (J - 1)dN_t$$

Answer key

The calibration instruments are short-dated smile and crash put quotes. Write a weighted least-squares objective using market quotes and model quotes.

Weights should reflect liquidity, bid-ask width, and quote reliability. Parameter constraints should prevent impossible prices or unstable extrapolation.

After calibration, inspect residuals and check whether each parameter has a distinct effect on the quotes.

$$\begin{aligned} \frac{dS_t}{S_t} &= (r - \lambda\kappa)dt + \sigma dW_t + (J - 1)dN_t \\ \hat{\theta} &= \arg \min_{\theta} \sum_i w_i (M_i(\theta) - Q_i^{\text{mkt}})^2 \end{aligned}$$

Question 17

For an equity option under Merton jump diffusion, compare the mathematical boundary condition with the trader's intuition. Give one limiting case where the answer should be obvious.

Answer key

The terminal condition comes from the payoff. Boundary conditions come from economic limits such as zero spot, very large spot, barrier touch, exercise, or discount-factor positivity.

For this product, the key boundary idea is compound Poisson distribution and terminal payoff.

A boundary condition is wrong if it permits arbitrage, negative values for non-negative payoffs, or a price that violates obvious monotonicity.

$$\begin{aligned} \frac{dS_t}{S_t} &= (r - \lambda\kappa)dt + \sigma dW_t + (J - 1)dN_t \\ V(T, S) &= g(S), \quad V(t, S_{\text{boundary}}) = \text{economic boundary value} \end{aligned}$$

Question 18

Design a convergence study for diffusion paths with random jump counts and jump sizes. Specify path counts, time-step refinement, confidence intervals, and the exact quantity plotted.

Answer key

Simulate diffusion paths with random jump counts and jump sizes. For each path, compute the payoff using only information available along that path.

The estimator is the discounted sample average. Report standard error and rerun with larger path counts or finer time steps to separate sampling error from discretization error.

For path-dependent products, store the running state variable explicitly.

$$\begin{aligned} \frac{dS_t}{S_t} &= (r - \lambda\kappa)dt + \sigma dW_t + (J - 1)dN_t \\ \hat{V}_0 &= e^{-rT} \frac{1}{N} \sum_{i=1}^N X^{(i)}, \quad \widehat{\text{SE}} = e^{-rT} \frac{s_X}{\sqrt{N}} \end{aligned}$$

Question 19

Find one way that forgetting the compensator and therefore shifting the forward price could pass a superficial unit test. Write a stronger test that would fail for the biased implementation.

$$\frac{dS_t}{S_t} = (r - \lambda\kappa)dt + \sigma dW_t + (J - 1)dN_t$$

Answer key

The main implementation risk is forgetting the compensator and therefore shifting the forward price. This can produce stable but wrong results because the error is systematic.

A good diagnostic compares the implementation with a limiting case or benchmark where the direction of the answer is known.

Change one input at a time and verify monotonicity, units, and convergence.

$$\begin{aligned}\frac{dS_t}{S_t} &= (r - \lambda\kappa)dt + \sigma dW_t + (J - 1)dN_t \\ \text{error}(N, \Delta t) &= \widehat{V}_{N, \Delta t} - V_{\text{benchmark}}\end{aligned}$$

Question 20

Explain an equity option under Merton jump diffusion and code the core routine. Give the derivation outline, pseudocode, and three numerical sanity checks.

Answer key

Create clear arrays for inputs, states, probabilities, discount factors, and payoffs. Check dimensions before combining them.

Return both the main value and diagnostics such as convergence, residuals, standard error, or benchmark difference.

The routine is acceptable only if the result can be reproduced from the displayed assumptions and data.

$$\begin{aligned}\frac{dS_t}{S_t} &= (r - \lambda\kappa)dt + \sigma dW_t + (J - 1)dN_t \\ \text{tree cost} &= O(n^2), \quad \text{simulation cost} = O(NM)\end{aligned}$$